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Recaptured image forensics based on local ternary count of high order prediction error



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ABSTRACT

With the advancement in image display technology and the widespread use of various image acquisition devices, recapturing high-quality images from high-fidelity LCD (liquid crystal display) screens becomes very convenient and easy. These recaptured images pose serious threats on current image forensic technology and intelligent recognition systems. In order to solve these security problems, we propose a recaptured image forensics method based on local ternary count (LTC) of high order prediction error. Specifically, given an image, the second-order difference maps on different directions are calculated in HSV color space for the original image and its average pooled version. Then, we calculate the prediction error maps from these second-order difference maps through a linear predictor. Finally, the normalized LTC histograms of all the downsampled versions of the obtained prediction error maps are calculated as features for training and testing. Single database experiments demonstrate that our proposed method performs very similar with the state-of-the-art traditional method on the most difficult-to-detect ICL-COMMSP database with obvious less execution-time and lower feature dimension. Meanwhile, it performs very close to the existing well-performed deep learning based method on all the databases, which verifies the effectiveness of our proposed features. Besides, mixed database experiments verify the superiority on generalization ability of our proposed method. Moreover, it is robust to JPEG compression.

1. Introduction

In the past few years, image display and acquisition technology have greatly flourished, which make it is convenient to recapture high-quality images from various display media, such as projection screens, printed materials, and LCD (liquid crystal display) screens. This process, which is known as image recapture, can be modeled in Fig. 1. In this figure, the left and the right dashed boxes present the processes of generating real-scene images and their corresponding recaptured images, respectively. Such recaptured images are so photorealistic that it is very difficult to detect by human eyes or even machines, which pose serious threats on image credibility and intelligent recognition systems.

Firstly, recapture operation can be utilized to evade image tampering detection, i.e., attackers can photograph the manipulated image by displaying it on a high-resolution LCD monitor to bypass the current image forensic methodologies [1–3] which are mainly based on searching statistical inconsistencies introduced by manual editing. As shown in Fig. 2, when tampered images undergo recapture operation, the obtained images are also "real images" directly produced by digital cameras as the specific artifacts introduced by manual operations such as resampling, splicing, copy–move can be eliminated automatically. Therefore, identifying recaptured image is an imperative task for researchers to optimize the current image forensics systems. Another threat brought by image recapture is rebroadcast attack, or also called spoofing. Nowadays, various intelligent recognition systems are widely used in our daily life. However, some of these systems are still vulnerable to rebroadcast attacks, i.e., attackers can fool some systems with a recaptured image of the original object. For instance, many supermarkets use automatic depositing bag systems to store customs' bags. Such system will give out a bar code scrip after the bag is stored and when a customer wants to take out his bag, he should put this bar code scrip before the scanner of the system to open the corresponding door of cabinet. However, if someone takes a photo of this bar code scrip, he may open the door for stealing. Therefore, it is very essential for an intelligent recognition system to own the ability of differentiating a real object or person from a recaptured image. In summary, it is significantly important to design reliable and automatic algorithms to identify high-quality recaptured images in our daily life.

In this paper, we address the problem of identifying the images recaptured from LCD screens. The motivation of our proposed method contains two aspects. On one hand, almost all the existing recaptured image detection (RID) approaches are mainly based on detecting some

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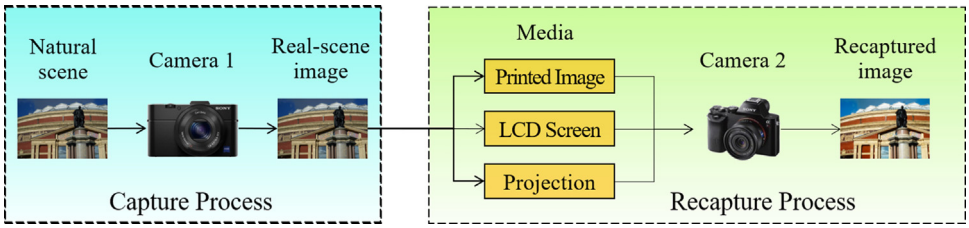


Fig. 1. The processes of generating real-scene images (left dashed box) and their corresponding recaptured images (right dashed box) [4].



Fig. 2. An illustration of some recaptured tampered images collected from NTU-ROSE database [5]. The first row presents some original images and the second row shows their corresponding recaptured tampered images. From left to right, the tampered images are created by color alteration, object insertion, and object removal, respectively.

specific artifacts introduced by recapture operation such as aliasing, color distortion, lighting environment inconsistencies, double JPEG compression, and edge blur artifact. However, some of these features may be invalid under some complicated recapture environment settings. For example, if the original image for recapturing is in PNG or BMP format, the double JPEG compression based RID methods will be invalid. Besides, shooting an image displayed in LCD screen in a darkroom can invalidate the RID methods based on environment lighting inconsistencies. By contrast, due to the shooting distances of real-scene images and recaptured images are usually different, the textures of these two kinds of images should be different. Therefore, texture descriptors, which make a weak assumption about the recapture environment settings, will be selected in this paper. Furthermore, given an image, we intend to extract features from its prediction error maps as some existing approaches [4,6,7] have verified that the features directly extracted from the original image in spatial domain would introduce redundancy. On the other hand, during recapture process, a new pixel interpolation operation will be introduced, which inevitably alters the correlations between R, G, and B channels. Therefore, there will exist differences in both brightness and color between real-scene images and their corresponding recaptured ones, which inspires us to extract features from HSV color space. Meanwhile, a multi-scale strategy which can present the variations of correlations better will be utilized. Single database experiments demonstrate that our proposed approach performs very similar with the state-of-the-art traditional method [7] on the most difficult-to-detect ICL-COMMSP database with obvious less execution-time and lower feature dimension, and very close to the other comparison traditional methods on NTU-ROSE and BJTU-IIS databases. Meanwhile, our proposed method performs very close to the existing well-performed deep learning based RID method on all the databases. On the other hand, mixed database experiments verify the superiority on generalization ability of our proposed method. Besides, experimental results verify the robustness of our proposed method to JPEG compression.

The remainder of this paper is organized as follows. The next section introduces the existing related work in this field. Section 3 describes our proposed approach in detail. Experimental results under different scenarios along with some useful discussions are given in Section 4. The last section concludes this paper and provides an outlook in the near future.

2. Related work

Recaptured image detection (RID) methodologies have received large attention in recent years and a growing number of literature can be found in this area. This task is aiming to decide a given image is real or recaptured, which is a typical binary classification problem. Almost all the existing approaches are based on the features aiming to analyze the statistical alterations in texture, edge, illumination, color, contrast, blurriness, noise, etc., which are introduced during image recapture process. According to the display media for image recapture, the existing approaches for RID can be divided into two categories: RID approaches for the images recaptured from (1) printed materials and (2) LCD screens.

In the early stage of RID, more attention was paid to detect the images recaptured from printed materials. As the pioneers, Yu et al. [8] found that the recaptured images and their corresponding original ones had different distributions on the gradient of the specular ratio, which inspired them to extract the micro-texture in printed paper from the specularity component of an image for RID. Based on the same finding, the work in [9] used two parameters to model the histogram of gradient on specularity component as features to distinguish recaptured facial images from real live ones. Gao et al. [10] proposed a physical model of image recapture and a set of features were inspired from this model. These features were composed of background contextual information, spatial distribution of specularity, surface gradient, color histogram and contrast. This method works well on low resolution recaptured images but the feature dimension is high. Later, in [11], a set of Markov process based features which extracted from the DCT coefficients array were utilized for RID. Zhai et al. [12] mentioned that the recapture process generally introduced some image quality losses, which motivated them to use texture features including local binary pattern variance and relative-contrast to identify the images recaptured from printed materials. This method works well on low-quality recaptured images. However, with the fast development of

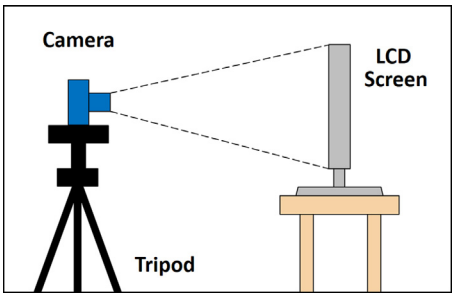


Fig. 3. Professional image recapture environment setting [4].

digital display technology and image acquisition devices, the quality of the images recaptured from LCD screens can be more satisfactory, especially under a professional image recapture environment setting (as shown in Fig. 3) whose parameters are determined by theoretical calculation [13]. As consequence, more and more attention has been paid to detect the images recaptured from LCD screens.

To detect the images recaptured from LCD screens, Cao et al. [5] designed several sets of features with 155-D including loss-of-detail features extracted with LBP (local binary patterns) [14] operator, color features reflecting the color anomalies, and texture features described by multi-scale wavelet statistics for detection. This approach runs very slow as the calculation of multi-scale LBP features is time-consuming. The work in [15] utilized noise features and compression features for RID. The authors first used a denoising algorithm to extract the noise of the given image and calculated the moment features of the obtained noise as noise features. Meanwhile, they applied the MBFDF (mode based first digit features) [16] to measure the double JPEG compression artifact as compression features. However, MBFDF only works for double JPEG compressed images, i.e., this algorithm requires both the real-scene image and the corresponding recaptured one must be in JPEG format, which not always be true in practice. Thongkamwitoon et al. [17] used a dictionary of edge profiles to detect recaptured images. Specifically, a dictionary of edge profiles was constructed corresponding to edges and then the maximum inner product was used to determine the best match between an edge found in a given image and a dictionary element. A crucial limitation of this method is the detection performance depends on the manual selected regions. The same group [18] then solved this problem by automatically extracting proper blocks owning a single sharp edge. Based on these selected blocks, they trained two dictionaries with K-SVD (K-singular value decomposition) and used the smallest error in sparse representation as feature for RID. Furthermore, they extended this work in [13] by using dictionary approximation error combined with mean edge spread width as features to identify high-quality recaptured images. In [19], the authors proposed two sets of features to describe the block and blurriness effect caused by double JPEG compression and the aliasing effect caused by unmatched frequency of screen and camera for RID. Then the same group [20] utilized quality aware features and difference matrix of DCT coefficients histogram features for RID. However, these two approaches [19,20] only work for JPEG compressed images. Based on the fact that recapture process usually introduced aliasing due to the high frequency period pattern of the screen pixel grid structure, the work in [21] and [22] used the 2D DFT of noise residual and the theory of cyclostationarity to identify recaptured images, respectively. However, according to [13], a professional setting of the recapture environment can suppress the aliasing artifact, which can decrease the performance of these two methods. Li et al. [7] proposed a RID method based on the dense local descriptors which had been used in the field of steganalysis [23]. However, the feature dimension is very high, making this method very time-consuming.

More recently, the work in [6] adopted the pixel-wise correlation coefficients in differential domain as features for RID. However, all the overlapping blocks are utilized to calculate correlation coefficients, which inevitably introduces some redundant information. In order to improve this approach, our previous work [4] calculated the enhanced residual-based correlation coefficients of some selected single-edge blocks to eliminate information redundancy. However, the selection of single-edge blocks is a little time-consuming and only small improvement is achieved when compared with [6]. The work in [24] used the features based on the pixel count of different edge types for RID. Specifically, the edges in an image were first classified into different categories with stationary wavelet transform and then the numbers of the edge pixels in different edge groups were extracted as feature. However, the number of edge pixels is varying due to various image contents, which may decrease the performance of this method. Sun et al. [25] used multi-scale wavelet statistics combined with uniform LBP of the Wiener-filtered version of the given image for RID and achieved promising performance. Then, the same group [26] brought up a new issue of "local planar linear point" to detect recaptured images. By establishing the mathematical model in imaging process, the concept and property of the local plane linear point could be derived from the model. Then the local plane linear point was extracted as feature for training and testing. More recently, the same group [27] extracted a 144-D gray level co-occurrence matrix (GLCM) feature from the wavelet transform subbands of an image for RID. However, experimental results have verified [25–27] perform not very well for the recaptured images in very high quality.

All the methods mentioned above used hand-crafted features to describe the statistical properties of recaptured images. In addition, some data-driven algorithms based on deep learning technology have also been reported. Yang et al. [28] proposed a promising method by adding an enhancement layer with Laplacian filter before CNN (convolutional neural networks) architecture to strengthen the differences in the features of real-scenes images and recaptured images for detection. However, a Laplacian filter with fixed parameters will ignore some other distortions in color and luminance. Choi et al. [29] proposed a CNN framework based on VGGNet [30] for detection. However, some false detection occurred in the blurred and flat regions, which decreases the performance due to block voting. Meanwhile, Li et al. [31] proposed a new framework by combining CNNs with RNNs (recurrent neural networks) for RID. The authors considered the convolutional operation as the preprocessing and the weights of the convolutional operator could be automatically learned during the training phase. Then, the features extracted from the trained CNN model were fed into a RNN to make classification. Although this method can achieve high performance, an inherent drawback of this hierarchical strategy is it requires higher hardware capacities. More recently, Anjum et al. [32] proposed a CNN architecture named HLReCapNet, which could identify both high-resolution and low-resolution recaptured images. However, for the images in different resolutions, this method used different pre-processing strategies. In other words, the performance would be affected by manual selection.

3. Proposed method

In this section, we present the whole process of LTC (local ternary count) features extraction,[1] which mainly comprises five components (as shown in Fig. 4): (1) color space conversion, (2) image average pooling, (3) second-order difference maps, (4) prediction error maps, and (5) normalized LTC histogram.

[1] The code of this paper can be downloaded from https://github.com/nanzhu-DMFLab/RID-SPIC-2022.

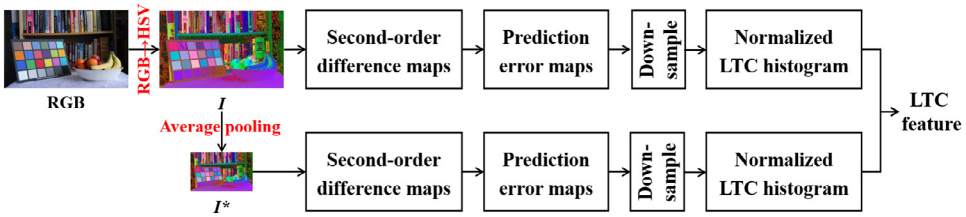


Fig. 4. LTC features extraction process.

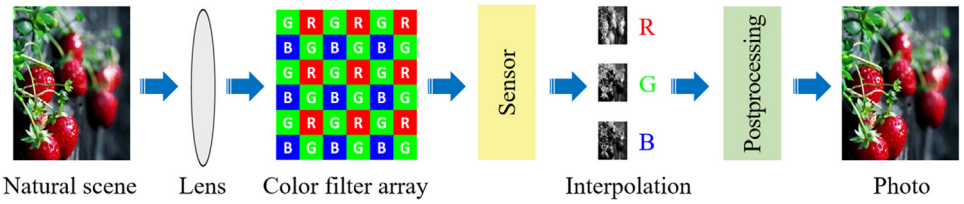


Fig. 5. Imaging process of a consumer camera.

3.1. Color space conversion

Although majority of the color distortions introduced during image recapture process can be eliminated with a professional environment setting, the color of these finely recaptured images still looks a little different from their original images (as shown in Fig. 9), which can be attributed to the correlations between different color channels have been altered. In detail, referring to Fig. 5, consumer cameras usually use a color filter array (CFA) to capture different color components and fill the missing RGB pixel values through the built-in interpolation algorithms to obtain three complete color channels. When filling the missing pixel values in a color channel, the state-of-the-art interpolation algorithms mainly use the pixel values in this channel, supplemented by the pixel values in the other two channels, which inevitably introduces correlations between R, G, and B channels. When photographing the obtained image again, an other interpolation process will be utilized. As a result, the correlations between different color channels will change. Intuitively, there existing differences in both brightness and color between real-scene images and their corresponding recaptured ones, so the features extracted in HSV color space can capture more distinct characteristics than in RGB color space. Therefore, given an image, we first convert it into HSV color space (denote the result as I, with size $M \times N \times 3$) for LTC features extraction.

3.2. Image average pooling

The normalized LTC histogram which will be used as the final LTC feature captures the correlations between image pixels in small scale. However, the correlations between pixels also exist in larger scale. In order to capture the correlations in a larger scale, we calculate the average pooled image I^* as

$$I^*(x,y,z) = \sum_{u=0}^{1}\sum_{v=0}^{1} \frac{I(2x-u, 2y-v, z)}{4}, \tag{1}$$

where $x \in \{1,2,\ldots,\lfloor M/2 \rfloor\}$, $y \in \{1,2,\ldots,\lfloor N/2 \rfloor\}$, $z \in \{1,2,3\}$, and $\lfloor\cdot\rfloor$ means floor operation.

As consequence, the size of the average pooled image I^* is 1/4 time of the original image I. In other word, two adjacent pixels in I^* can reflect the information of eight pixels in I. So the normalized LTC histogram of the average pooled image could express the information of the original image in a larger scale.

3.3. Second-order difference maps

Existing approaches [4,6,7] have proved that directly extracting features from the given image in spatial domain would introduce redundancy. Motivated by this, we try to extract the residual maps of an image and then calculate some statistics from these residual maps, which can reflect its subtle characteristics.

In order to capture the correlations between adjacent pixels, the second-order difference maps will be calculated. Using the hue channel of I (denoted as I_H) for example, let $I_H(i,j)$ represent the value at position (i,j), the second-order difference map in horizontal direction of I_H can be defined as

$$D^{(h)}_{I_H} = 2I_H(i,j) - I_H(i,j-1) - I_H(i,j+1). \tag{2}$$

Similarly, the second-order difference maps in vertical, diagonal and counter-diagonal directions of I_H can be defined as:

$$D^{(v)}_{I_H} = 2I_H(i,j) - I_H(i-1,j) - I_H(i+1,j), \tag{3}$$

$$D^{(d)}_{I_H} = 2I_H(i,j) - I_H(i-1,j-1) - I_H(i+1,j+1), \tag{4}$$

$$D^{(c)}_{I_H} = 2I_H(i,j) - I_H(i-1,j+1) - I_H(i+1,j-1). \tag{5}$$

For the saturation channel (I_S) and the value channel (I_V) of I, we calculate the second-order difference maps in the same manner. Meanwhile, for the average pooled image I^*, we also calculate its second-order difference maps in different directions.

3.4. Prediction error maps

Each of the aforementioned second-order difference maps can reflect the correlations between adjacent pixels in one direction. However, it overlooks the correlations between different directions. To further capture such correlations, we establish the prediction error maps through linear predictors of the second-order difference maps. Using the horizontal difference map of I_H as an example, the linear predictor for the magnitude of the second-order difference maps can be given by

$$|D^{(h)}_{I_H}| = w_1|D^{(v)}_{I_H}| + w_2|D^{(d)}_{I_H}| + w_3|D^{(c)}_{I_H}|, \tag{6}$$

where $|\cdot|$ means absolute value and w_k are weights. The matrix form of this equation can be formulated as $V_h = Q_h\vec{w}$, where V_h contains the value of $|D^{(h)}_{I_H}|$ dumped into a column vector. Q_h contains three columns which are the values of $|D^{(v)}_{I_H}|$, $|D^{(d)}_{I_H}|$ and $|D^{(c)}_{I_H}|$ at the same position as $|D^{(h)}_{I_H}|$ respectively, and $\vec{w} = (w_1, w_2, w_3)^T$. Here, to reduce the influence of noise, the magnitude in V_h and Q_h which are less than or equal to a defined threshold θ will be set as 0. As consequence, the best weights $\vec{w}$ can be obtained by

$$\vec{w} = (Q_h^T Q_h)^{-1} Q_h^T V_h. \tag{7}$$

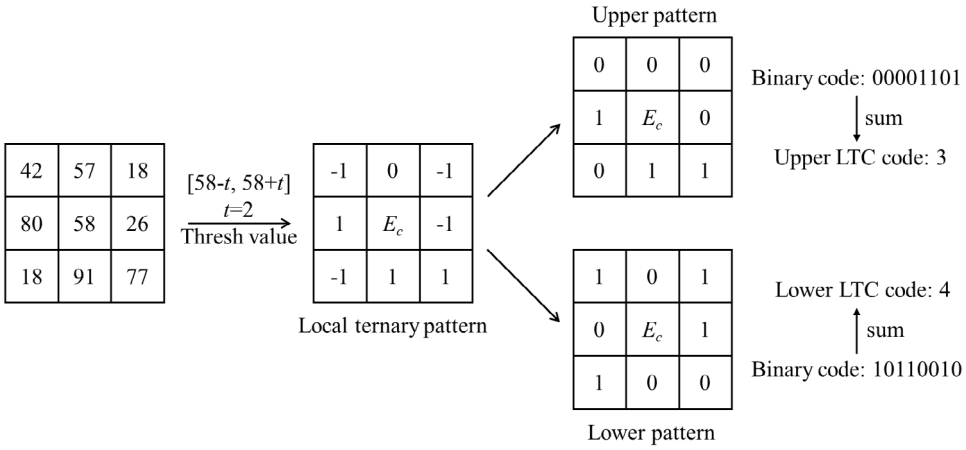


Fig. 6. The process of LTC code calculation.

After calculating $\vec{w}$, we restore both V_h and Q_h into 2-D matrix form. With the estimated predictor, the log error between the actual value and the predicted value in horizontal direction of I_H can be calculated as

$$P_{I_H}^{(h)} = \log_2(V_h) - \log_2(|Q_h\vec{w}|). \tag{8}$$

We call $P_{I_H}^{(h)}$ the prediction error map in horizontal direction of I_H. Similarly, the prediction error maps in other three directions can be acquired in the same manner. As consequence, we can get 4 prediction error maps as $P_{I_H}^{(*)}$, where $* \in \{h, v, d, c\}$ for I_H. Similarly, we can calculate all the prediction error maps in different directions for I^*. However, through experiments we find that there are some zeros in these obtained prediction error maps, which will lead to invalid LTC codes. In order to solve this problem, we downsample these prediction error maps with a fixed step λ. Specifically, we directly sample the pixels from each prediction error map from the top left to the bottom right with a fixed step λ without any other operations. The selection of the directions for calculating prediction error maps, the threshold θ used for noise removal, and the downsampling step λ will be discussed in Section 4.3. In the following, the normalized LTC histograms of the downsampled versions of these prediction error maps will be extracted to construct the final distinguishing features.

3.5. Normalized LTC histogram

The motivation of LTC features is mainly based on the differences in texture information between real-scene images and their corresponding recaptured ones. From this perspective, some texture descriptors will be considered. Although LBP descriptor has been proved to be highly discriminative feature for RID [5,25], it is sensitive to noise, particularly in near-uniform image regions as it utilizes the central pixel as threshold for binary coding directly. In order to solve this problem, Tan et al. extended LBP to a 3-value code as local ternary patterns (LTP) [33], which introduced threshold mode on the basis of LBP to further describe texture. Using Fig. 6 as an example, the indicator $f(E_i, E_c, t)$ is a 3-value function as

$$f(E_i, E_c, t) = \begin{cases} +1, & E_i > E_c + t \\ 0, & |E_i - E_c| \le t \\ -1, & E_i < E_c - t, \end{cases} \tag{9}$$

where E_c is the center pixel of this patch and E_i are the neighbor pixels of E_c. t is a user-specified threshold which is set to 2 in this example. Then, LTP splits the ternary pattern into an upper pattern and a lower pattern and treat them as two separate channels of LBP descriptors for binary coding.

Although LTP has higher discrimination ability and robustness than LBP, it dramatically increases the computational complexity and the feature dimension of LBP. In order to reduce computational complexity and maintain a relatively high performance, local ternary count (LTC) [34] was developed to roughly describe an image's local value distribution. The process of LTC extraction is similar to LTP but it reduces the LBP decimal coding process. Instead, as shown in Fig. 6, LTC performs bitwise sum of the binary code in the upper and lower patterns directly.

In summary, given a prediction error map $P^{(*)}$ with size $U \times V$, we first calculate the adjacent pixels' binary code according to (9) in 3×3 window and split the result into an upper pattern F_u and a lower pattern F_l. Secondly, the binary codes in the upper and lower patterns are accumulated to obtain the upper and lower LTC codes, respectively. Finally, the normalized LTC histogram $H(k)$ of all the F_u and F_l can be calculated as

$$H(k) = \frac{1}{U \times V} \sum_{p=1}^{U} \sum_{q=1}^{V} \ell(F_*(p, q), k), \tag{10}$$

where $k \in \{0, 1, \ldots, 8\}$ and $* \in \{u, l\}$. Function $\ell()$ outputs 1 if its two parameters are equal, otherwise it outputs 0. As consequence, for each prediction error map, a $9 \times 2 = 18$-dimension feature can be obtained. In summary, as we extract the normalized LTC histograms of a given image from 3 color channels (H, S, and V), 2 scales (original and average pooled), and N_p directions for calculating prediction error maps, the feature dimension of our proposed method is $3 \times 2 \times N_p \times 18 = 108 \times N_p$, where N_p will be selected in Section 4.3.

By the way, the limitation of LTC code calculation is it is difficult to set a suitable threshold as the prediction error maps are quite different. Here, we utilize a content-adaptive threshold as

$$t = \left| median(P^{(*)}) - \frac{1}{U \times V} \sum P^{(*)} \right|. \tag{11}$$

4. Experiments and discussions

In this section, we present the experimental results of our proposed approach on three public recaptured image databases in comparison with four latest traditional methods [6,7,25,27] and three deep leaning based methods [29,31,32] along with some useful discussions.

4.1. Databases

To evaluate the effectiveness of our proposed approach, we conduct a series of experiments on the following three public databases. All the images in these databases are acquired by different high-resolution DSLR (digital single lens reflex) cameras. The basic information which includes the number, format, size, and some distinct features of the images from these databases are described in the following. Figs. 7 to 9 present some representative image samples from these three databases



Fig. 7. Image samples from NTU-ROSE Database. The first row illustrates some real-scene images. The second row presents the corresponding recaptured versions.

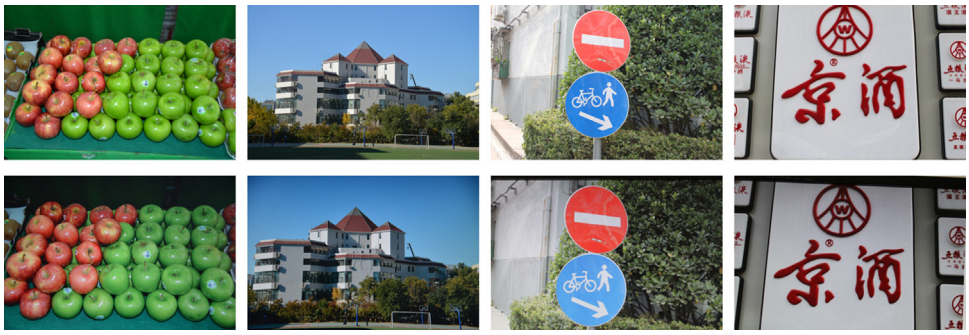


Fig. 8. Image samples from BJTU-IIS Database. The first row illustrates some real-scene images. The second row presents the corresponding recaptured versions.



Fig. 9. Image samples from ICL-COMMSP Database. The first row illustrates some real-scene images. The second row presents the corresponding recaptured versions.

respectively. For each figure, the first row illustrates some real-scene images and the second row presents the corresponding recaptured versions.

- **NTU-ROSE database**[2]: This database contains 2776 recaptured images with sizes [1536 × 2048 ~ 2448 × 3264] in JPEG format and 2710 real-scene images with sizes [600 × 800 ~ 3312 × 4416] in JPEG and BMP formats (only 100 real-scene images in BMP format). In this database, slight color distortion and misalignment are introduced during recapture process. Beside, slight aliasing artifacts exist in some recaptured images. Specifically, this database contains some recaptured tampered images which are visually very convincing (seeing the last two columns in Fig. 7 for example).
- **BJTU-IIS database** [19]: This database contains 706 recaptured images with sizes [2352 × 3136 ~ 4016 × 6016] in JPEG format and 636 real-scene images with sizes [2304 × 3456 ~ 4016 × 6016] in JPEG format. Some recaptured images in this database have black and/or white box(es) caused by misalignment (seeing the last two columns in Fig. 8 for example). Furthermore, slight aliasing artifacts can be found in some recaptured images from this database.
- **ICL-COMMSP database**[3]: This database contains 1440 recaptured images with sizes [1326 × 1778 ~ 1522 × 2286] in PNG format and 900 real-scene images with sizes [1932 × 2576 ~ 3648 × 5472] in JPEG format. The recaptured images in this database have good alignment but slight color distortions are introduced. Besides, the recaptured images in this database own the weakest aliasing artifacts when compared with the above two databases as the recapture environment is carefully set according to the parameters determined by theoretical calculation [13].

4.2. Experimental setting

In this paper, we conduct experiments on two scenarios. (1) Single database experiments. For each database, 20% of the real-scene images

[2] https://rose1.ntu.edu.sg/dataset/recapturedImages/.

[3] http://www.commsp.ee.ic.ac.uk/~pld/research/Rewind/Recapture/.

Table 1
Acc_All under different λ and θ (with prediction error maps under all the directions).

λ	$\theta = 1/255$	$\theta = 2/255$	$\theta = 3/255$	$\theta = 4/255$	$\theta = 5/255$	$\theta = 6/255$
1	0.9413	0.9515	0.9551	0.9538	0.9523	0.9519
2	**0.9624**	**0.9653**	**0.9641**	**0.9628**	**0.9617**	**0.9608**
3	0.9375	0.9464	0.9489	0.9504	0.9503	0.9483

and the recaptured images are used for training while all the rest images are used for testing. (2) Mixed database experiments. 20% of the real-scene images and the recaptured images from each database are combinedly used for training while all the rest images from these three databases are used for testing. For the deep learning based methods [29,31,32], we follow the experimental setting used by the authors but use the same input size 96×96.

Unlike the work in [6,13] rescaled all the images into a fixed width of 2048 pixels for training and testing or even manually selected images for training, in this paper, as fair as possible, we randomly select images for training without any additional operations. LIBSVM [35] with RBF (radial basis function) kernel is utilized as classifier for all the traditional methods and the hyper-parameters of this classifier are determined by the built-in tool "easy.py". By the way, the work in [7] tested two strategies including (1) 988-D feature + SVM classifier, and (2) 3276-D feature + ensemble classifier [36]. The feature extraction process in the second strategy is extremely time-consuming and the feature is too large for SVM classifier to carry out training. By taking both fairness and execution time into consideration, we use the first strategy as it uses LIBSVM as classifier like all the other traditional methods. In order to make comparison conveniently, we use the metrics defined in the comparison methods to evaluate the performance. Specifically, by regarding the recaptured images as positive samples (P) and the real-scene images as negative ones (N), the metrics *Acc_All*, *Acc_Rec.*, and *Acc_Real* used in this paper can be defined as

$$Acc_All = \frac{TP + TN}{P + N}, \tag{12}$$

$$Acc_Rec. = TP/P, \tag{13}$$

$$Acc_Real = TN/N, \tag{14}$$

where TP and TN stand for the correctly detected recaptured images and real-scene images, respectively. Finally, for all the scenarios, we use the mean value obtained from 50 runs as the final result.

4.3. Parameter selection

In this subsection, the directions for calculating prediction error maps, the threshold θ used for noise removal, and the downsampling step λ will be selected by experiments. Here, we use the most high-quality ICL-COMMSP database for parameter selection. Firstly, we use the prediction error maps under all the directions to decide the downsampling step λ. Table 1 presents the overall accuracy under different θ and λ. The largest value under each θ is in bold. The reason why we used $\theta \in [1/255, 6/255]$ is the noise in an image is usually very small (the pixel values in an image are normalized into [0,1]). From this table we can easily find that the proposed method can always achieve the best *Acc_All* under $\lambda = 2$. This phenomenon can be attributed to when $\lambda = 1$, i.e., without downsampling, the zeros in the prediction error maps will introduce invalid LTC codes. On the other hand, a large downsampling step will eliminate some useful information, which inevitably decreases the correlations between pixels. Therefore, we use $\lambda = 2$ throughout this paper. As consequence, the sizes of the prediction error maps will decrease as one fourth of the original sizes.

In addition, as the calculation of prediction error maps is a little time-consuming, we intend to select the directions for calculating the final features. Table 2 presents the overall accuracy under different θ and N_p with $\lambda = 2$. Here, $N_p = 1$ means only the prediction error map in horizontal direction is used and $N_p = 2$ means the prediction error maps in both horizontal direction and vertical direction are used, and so on until to $N_p = 4$. The largest value under different N_p is in bold. From this table we can find that the results under $N_p = 3$ is close to the results under $N_p = 4$ and is obviously higher than the results under $N_p = 2$, especially for $\theta = 2/255$ and $\theta = 3/255$, which can be attributed to the symmetry of diagonal and counter-diagonal directions. By taking both accuracy and execution time into account, we use $N_p = 3$ and $\theta = 3/255$ throughout all the experiments. As consequence, the final feature dimension of the proposed method is $108 \times N_p = 324$.

Table 2
Acc_All under different N_p and θ ($\lambda = 2$).

θ	$N_p = 1$	$N_p = 2$	$N_p = 3$	$N_p = 4$
1/255	0.9312	0.9425	0.9596	0.9624
2/255	0.9416	0.9495	0.9614	**0.9653**
3/255	**0.9457**	**0.9539**	**0.9622**	0.9641
4/255	0.9445	0.9504	0.9589	0.9628
5/255	0.9398	0.9501	0.9568	0.9617
6/255	0.9342	0.9483	0.9557	0.9608

4.4. Single database experiments

Tables 3 to 5 present the comparison results on NTU-ROSE, BJTU-IIS, and ICL-COMMSP databases, respectively. The maximum value under each metric is bold. Here, in order to train CNNs, we use data augmentation such as overlapped cropping, rotation, and flipping for the deep learning based methods [29,31,32]. From these tables we can find that almost all the methods can achieve *Acc_All* more than 97% on NTU-ROSE and BJTU-IIS databases as the quality of the recaptured images from these two databases is not high enough. Specifically, although we use ICL-COMMSP database for parameters selection, our results on NTU-ROSE and BJTU-IIS databases are very close to the results of all the comparison methods. In comparison, for the recaptured images from ICL-COMMSP database, our proposed method and [7] perform at the same level and outperform other three traditional methods obviously. The reason why the proposed method performs well on ICL-COMMSP database but very close to or even a little less than some traditional methods on NTU-ROSE and BJTU-IIS databases can be attributed to two aspects. On one hand, we utilize ICL-COMMSP database to decide some parameters of the proposed method, which will not be the optimal strategy to detect the images from the other two databases. The same phenomenon also exist in [7], in which the authors used ICL-COMMSP database for feature selection. On the other hand, our proposed method and [7] extract features from high order residual maps, which are more suitable to describe the subtle alterations introduced in high-quality recaptured images. Meanwhile, our method performs very close to the state-of-the-art deep learning based method [31], which verifies the effectiveness of the proposed LTC features. Besides, the deep learning based method [29] usually achieves a better *Acc_Real* but a lower *Acc_Rec.*, which leads to a relatively lower *Acc_All*. This phenomenon can be attributed to this algorithm is based on block voting, but the flat or blurred blocks are usually incorrectly classified as real ones. The deep learning based methods [31] and [32] perform relatively worse on BJTU-IIS database as the images in this database are a little not enough for training CNNs although we use data augmentation. In contrast, [29] uses deep transfer learning, which needs relatively less training samples, receiving higher performance than [31] and [32] on BJTU-IIS database. However, when the training samples are enough (for NUT-ROSE database and ICL-COMMSP database), [31] performs better, which can be attributed to this method uses RNN to excavate the correlations between image patches.

Table 3
Detection results on NTU-ROSE database.

Method	*Acc_All*	*Acc_Rec.*	*Acc_Real*
[6]	0.9784	0.9802	0.9766
[7]	0.9752	0.9815	0.9687
[25]	0.9773	0.9756	0.9790
[27]	0.9710	0.9743	0.9677
[29]	0.9764	0.9725	**0.9803**
[31]	**0.9785**	**0.9833**	0.9734
[32]	0.9766	0.9807	0.9724
Proposed	0.9744	0.9813	0.9672

Table 4
Detection results on BJTU-IIS database.

Method	*Acc_All*	*Acc_Rec.*	*Acc_Real*
[6]	0.9778	**0.9879**	0.9665
[7]	0.9746	0.9803	0.9683
[25]	**0.9800**	0.9833	0.9763
[27]	0.9747	0.9796	0.9693
[29]	0.9768	0.9731	**0.9809**
[31]	0.9712	0.9751	0.9669
[32]	0.9633	0.9641	0.9624
Proposed	0.9752	0.9828	0.9670

Table 5
Detection results on ICL-COMMSP database.

Method	*Acc_All*	*Acc_Rec.*	*Acc_Real*
[6]	0.9224	0.9423	0.8907
[7]	0.9617	0.9725	0.9444
[25]	0.9157	0.9449	0.8689
[27]	0.9213	0.9414	0.8891
[29]	0.9240	0.9257	0.9213
[31]	**0.9632**	0.9727	**0.9478**
[32]	0.9490	0.9567	0.9366
Proposed	0.9622	**0.9759**	0.9403

Table 6
Detection results on mixed database.

Method	*Acc_All*	*Acc_Rec.*	*Acc_Real*
[6]	0.9304	0.9482	0.9098
[7]	0.9489	0.9593	0.9368
[25]	0.9289	0.9437	0.9117
[27]	0.9222	0.9277	0.9158
[29]	0.9285	0.9268	0.9304
[31]	0.9487	0.9565	**0.9394**
[32]	0.9394	0.9521	0.9246
Proposed	**0.9526**	**0.9654**	0.9376

4.5. Mixed database experiments

In addition, we conduct mixed database experiments and present the results on Table 6. From this table we can find that for all the methods, the results obtained from the mixed database are worse than the results on NTU-ROSE and BJTU-IIS databases, which can be attributed to the diversity of the images in the mixed database has increased. In other words, the quality of the images in the mixed database varies a little large. However, our proposed method outperforms the others, especially on *Acc_All* and *Acc_Rec.*, which demonstrates the superiority of the proposed LTC features again. Besides, for our proposed method, the performance obtained from each database is more similar with the performance achieved on the mixed database, i.e., it has good generalization ability. By the way, the improvement of the performance on the mixed database is very significant as the images given to detect in practice can be very diverse.

4.6. Effectiveness verification of multi-scale strategy

In this subsection, we test the effectiveness of the proposed multi-scale strategy. Table 7 presents the detection results of the LTC features of the images in original scale and the average pooled scale, respectively. Here, "avgpool" is short for average pooled. From this table we can find that the LTC features extracted from the images in original scale can be more effective, which can be attributed to the left artifact is more obvious in original image sizes. Besides, when compared with the detection results in single database experiments, improvement can be achieved by fusing the LTC features of the images in two different scales, which verifies the effectiveness of our proposed multi-scale strategy. However, by fusing the LTC features in two scales, *Acc_All* can be improved about 5% for ICL-COMMSP database, while less than 1% growth can be achieved in the other two databases. In order to explain this phenomenon, we show the average feature differences between real-scene images and recaptured images in two scales in ICL-COMMSP database (c.f. Fig. 10) and BJTU-IIS database (c.f. Fig. 11), respectively. From Fig. 10 we can find that the average feature differences are quite different for the images in original scale and the average pooled scale (as shown in the green dotted circles and red solid circles). In contrast, the distributions of the average feature differences of the two scales in BJTU-IIS database are more similar. Therefore, the LTC features in two different scales perform similar in this database. As summary, we can conclude that our proposed multi-scale strategy can improve performance and is especially effective for high-quality recaptured images.

Table 7
Detection results of the LTC features of the images in original scale and the average pooled scale.

Database	Scale	*Acc_All*	*Acc_Rec.*	*Acc_Real*
NTU-ROSE	Original	0.9654	0.9698	0.9610
	Avgpool	0.9503	0.9599	0.9405
BJTU-IIS	Original	0.9688	0.9742	0.9628
	Avgpool	0.9563	0.9650	0.9466
ICL-COMMSP	Original	0.9158	0.9426	0.8728
	Avgpool	0.8871	0.9181	0.8374

4.7. Computational complexity

In this subsection, we use CPU time to compare the computational complexity between all the traditional methods. Table 8 presents the average execution time of feature extraction for the images with size $s \times s$ from all the databases (for the image whose size is larger than the required size, we select the central portion with the required size. For the image whose size is smaller than the required size, we pad the image with mirror reflections of itself along the right and post edges). The feature dimensions of [6,7,25,27], and the proposed method are 28, 988, 89, 144, and 324, respectively. The results are obtained by Matlab 2014b (desktop PC with a 3.0 GHz Core i5-7400 processor and 8 GB RAM).

From this table we can find that the execution time of all the methods increase with the image size. Specifically, [6,25,27] run at the same level and faster than our proposed method. When the image size is small, our proposed method runs relatively fast. However, with image size increasing, the execution time of our proposed method increases faster than [6,25,27], which can be attributed to the execution time for calculating prediction error maps increases significantly. In comparison, [7] runs very slow due to its complex calculation process and high feature dimension.

4.8. Robustness testing

In practice, recapture operation is usually used as the last step of image tampering to bypass the existing image forensics systems. However, almost all the social websites use JPEG compression for the uploaded images. Therefore, we should test the robustness of our proposed method to JPEG compression. Here, we use the recaptured

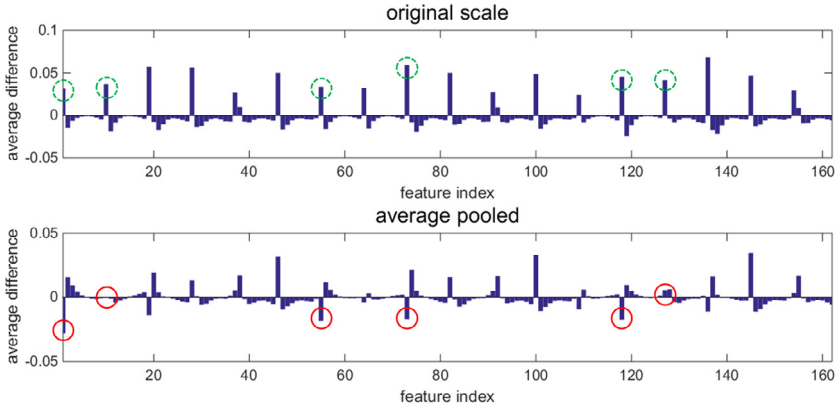


Fig. 10. Average feature differences between real-scene images and recaptured images in different scales in ICL-COMMSP database.

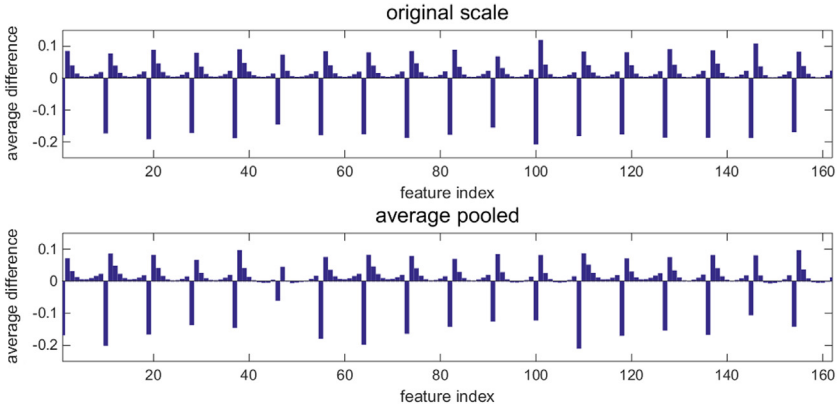


Fig. 11. Average feature differences between real-scene images and recaptured images in different scales in BJTU-IIS database.

Table 8
Average execution time (seconds) under different image sizes.

Method	$s = 512$	$s = 768$	$s = 1024$	$s = 2048$	$s = 3072$
[6]	0.14	0.32	0.54	2.48	5.42
[7]	5.32	12.01	20.79	81.07	185.45
[25]	0.38	0.47	0.64	2.57	5.45
[27]	0.12	0.31	0.58	2.58	5.83
Proposed	0.36	0.86	1.58	6.53	13.24

images in ICL-COMMSP database to verify the robustness to JPEG compression as only the recaptured images in this database are PNG format, while all the other images in the mentioned three databases have already been JPEG compressed with different quality factors. In detail, we use 20% of the real-scene images and the recaptured images from ICL-COMMSP database for training and use the compressed version of all the recaptured images in this database for testing.

Table 9 demonstrates the *Acc_Rec*. of our proposed method under different JPEG compression quality factors Q (we use Matlab "imwrite" function to compress images). As we expected, the performance of our proposed method decreases with a diminishing Q as JEPG is lossy compression. However, even though $Q = 60$, the detection accuracy of our proposed method is still larger than 85%, which verifies the robustness of our proposed method to JPEG compression.

5. Conclusion

In this paper, motivated by the different imaging mechanism and processes of real-scene images and recaptured ones, we propose a simple yet effective recaptured image detection method based on the normalized LTC histograms of high order prediction error maps. Single database experiments demonstrate that our proposed approach performs very similar with the state-of-the-art traditional method [7] on the most difficult-to-detect ICL-COMMSP database with obvious less execution-time and lower feature dimension. Except [7], the proposed method outperforms the other traditional comparison methods obviously on high-quality recaptured images and performs very close to them on relative low-quality recaptured images from NTU-ROSE and BJTU-IIS databases, which verifies the superiority of our method on the ability of describing subtle alterations introduced in high-quality recaptured images. Meanwhile, the proposed method performs very close to the existing well-performed deep learning based RID method [31] on all the databases, but it does not need high hardware capacities. On the other hand, mixed database experiments verify the superiority of our proposed method on generalization ability. Besides, it is robust to post JPEG compression. However, our method still has a large room for improvement. Firstly, we will attempt to design more efficient prediction error maps calculation strategies to accelerate our proposed method and improve its performance. Besides, some novel texture descriptors [37,38] and color features [39,40] will be considered to further improve the performance.

Table 9
Acc_Rec. of our proposed method under JPEG compression.

Quality	Q=95	Q=90	Q=80	Q=70	Q=60
Acc_Rec.	0.9345	0.9120	0.8952	0.8876	0.8523

CRediT authorship contribution statement

Nan Zhu: Conceptualization, Methodology, Software, Writing – original draft. **Zhiqin Liu:** Data curation, Software, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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